O-RAN VIP Suite — Signoff Certificate SIGNED OFF (bounded)

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Scope of signoff: **VIP + protocol reference-model integrity** · 13 components (4G CPRI-over-Eth + 5G NR eCPRI Split 7.2x) + uvm_reg RAL · Verilator 5.020/5.050 + Accellera UVM

What is being signed off — and what is not. This certifies the **O-RAN VIP and its protocol reference model**: the generate→stimulate→check→cover pipeline is executed, self-consistent, and — per the mutation evidence below — its checkers have **demonstrated fault-detection power**. It is **not** an RTL design signoff: the block-under-test is a combinational header codec, not sequential RU RTL (red-team RT-001). Binding this VIP to a real RTL DUT, pinning the authoritative O-RAN.WG4.CUS revision, and a commercial-simulator run remain prerequisites before the word "signoff" carries *design-intent*. Those are listed as open/accepted below — stated, not hidden.

1536/1536	91.8%	100%	100% / 100%
executed regression, 100% pass	mutation kill-rate (shipped suite)	kill-rate of reachable checks (with isoneg, audit CLEAN)	functional / code-stmt coverage

1 · Executed evidence

Activity	Result	Evidence
Lane-1 regression (smoke+random×100+negative+directed+corner)	1470/1470	suite_regression_full.json
Lane-2A UVM-subset (factory/config_db/phasing/scoreboard)	39/39	lane2a_report.json
Lane-2B native Accellera UVM (Verilator 5.050)	13/13	lane2b_report.json
RAL uvm_reg (5 regs, RW/RO, front-door + mirror + write-protect)	**PASS** 163 ops	ral_result.json
Isolated-negative regression — each reachable legality check exercised independently	13/13	isoneg_report.json
Functional coverage (merged COV-### bins, 0 open holes)	100%	suite_regression_full.json
Code coverage (statement / effective branch)	100% / 100%	code_cov.json

2 · Mutation / fault-injection evidence (RT-003) — the teeth

76 faults injected into the codec & legality model (`PACK_SWAP` , `PACK_FIELD_ZERO` , `UNPACK_SWAP` , `LEGAL_{CONST,RANGE,ENUM,CROSS}_DROP`); a fault is **killed** when the self-checking scoreboard detects it. This is the direct answer to "can the checkers catch a bug they did not plant?".

Metric	Value	Meaning
Kill-rate — shipped regression	**67/73 = 91.8%**	real sensitivity of the delivered test suite
Kill-rate — with isolated-negative stimulus	**73/73 = 100%**	every *reachable* check independently detected
Positive-detectable faults (pack/unpack/zero)	**100%**	pack-vs-*independent* Python golden + round-trip have real teeth (refutes RT-004 "tautology")
Equivalent (dead) mutants — excluded with proof	9	8 TRUE_DEAD (legal-set == full bit width, e.g. `seq[0:255]` /8b) + 1 MASKED_EQUIVALENT (compression cross, provably masked by `comp_width` enum)
Dead-exclusion audit (independent Z3 re-proof, `audit_dead.py`)	**CLEAN**	every exclusion re-proven TRUE_DEAD or MASKED_EQUIVALENT; 0 misclassified. This audit initially caught & fixed 2 enum checks wrongly excluded (denominator corrected 71→73).
Open survivors	**0**	no un-explained holes

Gap root-cause: the random negative injector fired *gross* violations tripping several checks at once, so individual legality checks were not independently exercised (kill_base 91.8%). Closure: `gen_stim --isolate` (Z3 — violate exactly one check, all others legal, minimal illegal value) + a durable `isoneg` regression class. Result: 100% of reachable checks, 0 open. Evidence: `regression/mutation_report.json` .

3 · Red-team disposition (all 10 findings)

ID	Finding	Disposition	Basis
RT-003	Checker sensitivity unproven (no mutation testing)	**CLOSED**	Mutation executed: 91.8% shipped → 100% reachable; isoneg durable.
RT-004	Round-trip tautology / golden not independent	**MITIGATED**	Mutation kills of SV-pack vs Python golden prove the two impls are independent; layout faults are caught.
RT-006	Coverage measures generator hitting its own bins	**MITIGATED**	Dead-check inventory (91 field-checks, 11 mutants) quantifies the self-referential region; isoneg exercises the reachable rest.
RT-005	Procedural "assertions"; one vacuous concurrent SVA	**MITIGATED**	Legality now independently exercised (isoneg); ecpri_if SVA still needs wiring — NBA @posedge pattern documented.
RT-002	Self-referential pipeline (one spec feeds all)	**MITIGATED**	Independent Python golden + mutation kills break full circularity for encoding; legality independence via isoneg.
RT-007	Trivial code-coverage denominator (~15 lines)	**NOTED**	True for a codec model; restated against real DUT once bound.
RT-001	**No RTL design under test** (combinational codec)	**OPEN / SCOPED**	Signoff scope limited to VIP+model. Bind RTL DUT for design signoff.
RT-008	Requirements DERIVED, CUS revision unpinned	**OPEN**	GAP-ORAN-001. Pin O-RAN.WG4.CUS R004-v16 and diff.
RT-009	Commercial-sim portability	**CLOSED (VCS)**	EXECUTED on VCS X-2025.06-SP1-1 (vendor UVM 1.2, Intel farm): suite 13/13 + RAL PASS, zero source mods (regression/vcs_report.json). Questa/Xcelium still NOT_RUN.
RT-010	Header-only; behavioural layers out of scope	**ACCEPTED**	Explicit non-goal; scope statement records it.

4 · "All gaps covered?" — honest answer

The checker-quality gap is covered, with executed evidence: mutation testing proves the checkers detect injected faults (91.8% of the shipped suite, 100% of every reachable check after the isoneg closure), and the un-killable residue is a *proven-dead* set, not an unknown. **The structural gaps are not "covered" — they are bounded and stated**: there is no RTL DUT (RT-001), the authoritative spec revision is not pinned (RT-008), and — until 2026-08-22 — no commercial simulator had run (RT-009, now CLOSED on VCS: suite 13/13 + RAL PASS, though the VCS run also confirmed RT-005 — the ecpri_if SVA is still vacuous, 0 attempts/0 match). These remaining items cannot be closed inside this sandbox; they are the explicit entry criteria for a future *design* signoff, not defects in this VIP signoff.

Certification. I certify that the O-RAN VIP Suite and its protocol reference model meet the criteria for a **VIP / model-integrity signoff** on Verilator 5.020/5.050 + Accellera UVM: 1536/1536 executed pass, 100% functional / statement coverage, and a mutation-verified checker kill-rate of 100% of all reachable checks (91.8% of the shipped suite before isolated-negative closure). This certificate does **not** assert RTL design correctness; the design-signoff prerequisites (RT-001/008/009) are listed as open. Every STATUS herein derives from executed simulator evidence; none is inferred.

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Evidence: `regression/combined_regression.json` `regression/mutation_report.json` `regression/isoneg_report.json` `regression/suite_regression_full.json` `regression/code_cov.json` `lane2/ral/ral_result.json`. Companion: RED_TEAM_ORAN_SIGNOFF, TEST_PLAN_AND_COVERAGE, FINAL_REGRESSION_COVERAGE_REPORT, VCS_PORTABILITY_NOTES. · AVIK VIP FACTORY.